Hierarchical modelling

Bayesian multilevel models

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Outline

- Models characterized by several parameters
- Assumptions of linear models
- What we want beyond linear model
- Hierarchical models and levels of hierarchy
- Comparing to frequenist random effects
- Bayesian advantages
- Model selection
- Information criteria

Models characterized by several parameters

- In many statistical applications, the model is characterized by several parameters.
- Many real-world problems involve data grouped by source, area, time, etc.
- Non-hierarchical models assume independence between units - often inappropriate.

Models characterized by several parameters

- Sample `R` code:

```r
data <- read_rds(here("data", "italy", "italy_mortality.rds"))
glimpse(data)
summary(data)
```

Models characterized by several parameters

```r
data <- read_rds(here("data", "italy", "italy_mortality.rds"))
glimpse(data)
```

```
Rows: 96,300
Columns: 13
$ NOME_PROVINCIA          <chr> "Agrigento", "Agrigento", "Agrigento", "Agrige~
$ SIGLA                   <chr> "AG", "AG", "AG", "AG", "AG", "AG", "AG", "AG"~
$ date                    <date> 2013-05-04, 2013-05-05, 2013-05-06, 2013-05-
0~
$ deaths                  <dbl> 11, 11, 12, 8, 10, 9, 14, 14, 5, 12, 10, 11, 9~
$ year                    <dbl> 2013, 2013, 2013, 2013, 2013, 2013, 2013, 2013~
$ expected                <dbl> 11.75367, 11.75367, 11.75367, 11.75367, 11.753~
$ hol                     <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0~
$ month                   <dbl> 5, 5, 5, 5, 5, 5, 5, 5, 5, 5, 5, 5, 5, 5, 5, 5~
$ temperature_lag0_3      <dbl> 20.47926, 20.25625, 20.00277, 18.80698, 17.427~
$ relativehumidity_lag0_3 <dbl> 41.23675, 49.12295, 54.27453, 56.57282, 63.886~
$ o3_lag0_3               <dbl> 101.69444, 98.83333, 99.25000, 100.05556, 104.~
$ pm10_lag0_3             <dbl> 40.13326, 39.26412, 35.22450, 29.67259, 26.707~
$ week                    <dbl> 18, 18, 18, 19, 19, 19, 19, 19, 19, 19, 20, 20~
```

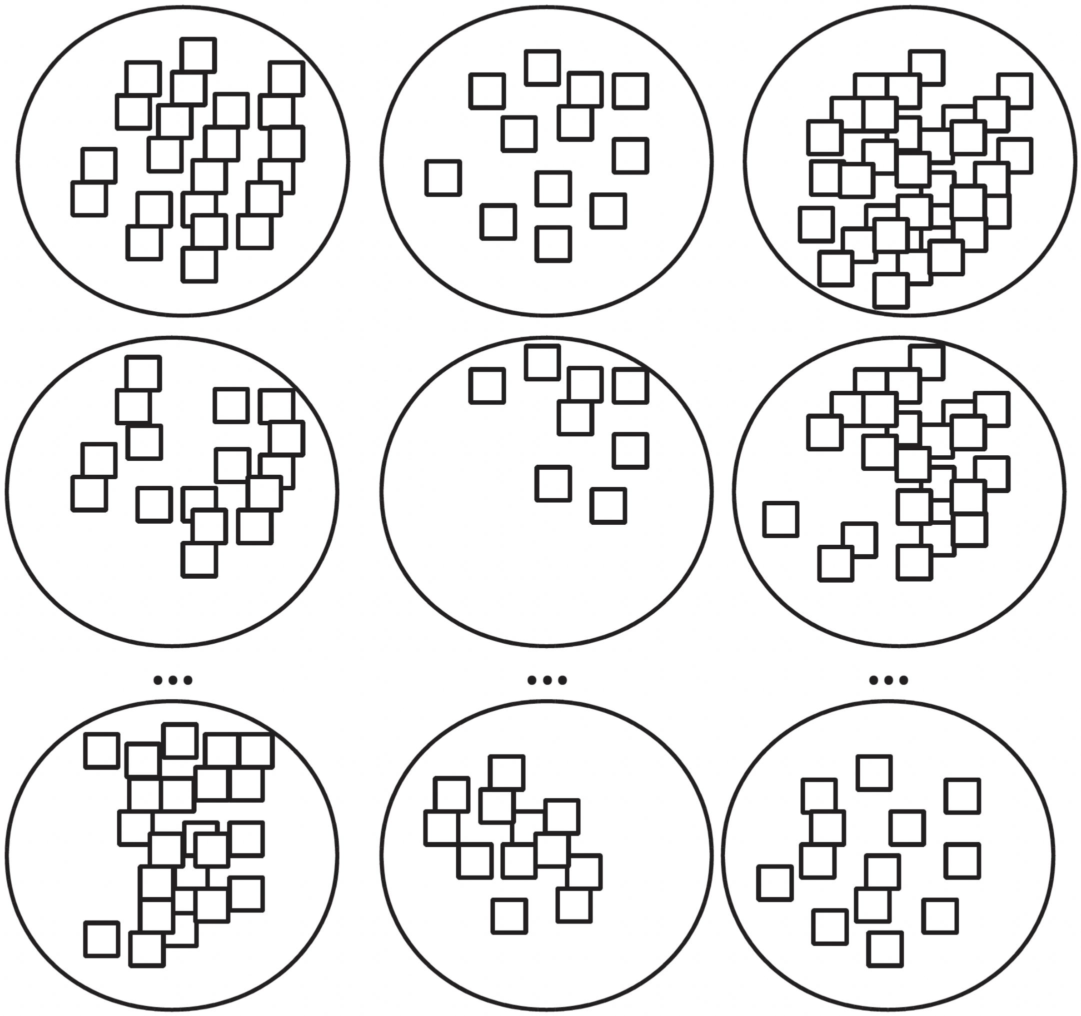
Models characterized by several parameters

```r
summary(data)
```

```
   NOME_PROVINCIA        SIGLA                 date                deaths      
 Length   :96300    Length   :96300    Min.   :2013-05-04   Min.   :  0.00  
 N.unique :  107    N.unique :  107    1st Qu.:2014-07-17   1st Qu.:  7.00  
 N.blank  :    0    N.blank  :    0    Median :2016-01-16   Median : 11.00  
 Min.nchar:    4    Min.nchar:    2    Mean   :2016-01-16   Mean   : 14.93  
 Max.nchar:   21    Max.nchar:    2    3rd Qu.:2017-07-17   3rd Qu.: 17.00  
                                       Max.   :2018-09-30   Max.   :272.00  
      year         expected            hol               month      
 Min.   :2013   Min.   :  2.909   Min.   :0.00000   Min.   :5.000  
 1st Qu.:2014   1st Qu.:  8.231   1st Qu.:0.00000   1st Qu.:6.000  
 Median :2016   Median : 11.674   Median :0.00000   Median :7.000  
 Mean   :2016   Mean   : 16.223   Mean   :0.01333   Mean   :7.033  
 3rd Qu.:2017   3rd Qu.: 18.153   3rd Qu.:0.00000   3rd Qu.:8.000  
```

```
Max.   :2018   Max.   :118.261   Max.   :1.00000   Max.   :9.000
temperature_lag0_3 relativehumidity_lag0_3   o3_lag0_3       pm10_lag0_3
Min.   :-2.582     Min.   :24.01           Min.   :  1.00   Min.   : 2.917
1st Qu.:17.029     1st Qu.:63.92           1st Qu.: 96.06   1st Qu.:14.375
Median :20.380     Median :71.47           Median :108.38   Median :18.333
Mean   :19.996     Mean   :70.76           Mean   :111.35   Mean   :19.200
3rd Qu.:23.346     3rd Qu.:78.49           3rd Qu.:123.50   3rd Qu.:22.875
Max.   :32.962     Max.   :96.78           Max.   :257.50   Max.   :92.226
     week
Min.   :18.0
1st Qu.:23.0
Median :29.0
Mean   :28.8
3rd Qu.:34.0
Max.   :40.0
```

Models characterized by several parameters



Models characterized by several parameters

Models characterized by several parameters

Assumptions of linear models

- Independence of observations.
- Constant error variance (homoskedasticity).
- Single set of parameters apply to all observations.
- No measurement error in predictors.
- Complete data.

Assumptions of linear models

- Independence of observations. Data often clustered.
- Constant error variance (homoskedasticity). Variability often differs across groups.
- Single set of parameters apply to all observations. Subgroups may have different parameters.
- No measurement error in predictors. Predictors are often noisy.
- Complete data. Missingness is common.

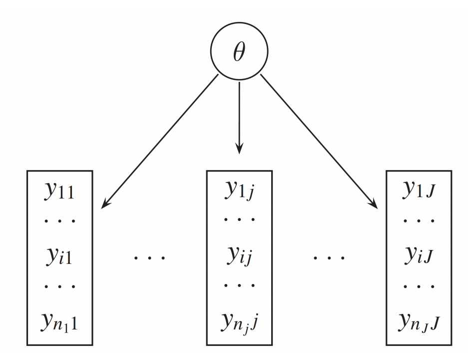
What we want beyond linear model

- To appropriately model grouped or nested data (e.g., individuals within regions, time within subjects).
- To allow for group-specific effects (e.g., varying intercepts or slopes across groups).
- To borrow strength across groups, improving estimates for small or noisy groups.
- To relax the assumption of independence by accounting for intra-group correlation.
- To model heterogeneity in variances across groups or levels.
- To introduce structure in parameters, linking them through shared hyperparameters.

What we want beyond linear model

- To propagate uncertainty across levels of the model (data, parameters, hyperparameters).
- To handle missing data naturally through the model hierarchy.
- To incorporate prior knowledge at multiple levels of the model.
- To support flexible generalizations (e.g., nonlinear relationships, spatiotemporal structure, GLMs).
- To incorporate prior information and quantify uncertainty at multiple levels.
- To represent spatial, temporal, or hierarchical dependence in data.

Complete pooling



- A *pooled* model implies that the data are sampled from the same model.
- This ignores all variation among the units being sampled.
- All observations share common parameter θ.
- It ignores any differences between measurement blocks and does not acknowledge variability block to block (second level units).
- However, observations within each unit are more likely to be like each other.

Complete pooling (from lab)

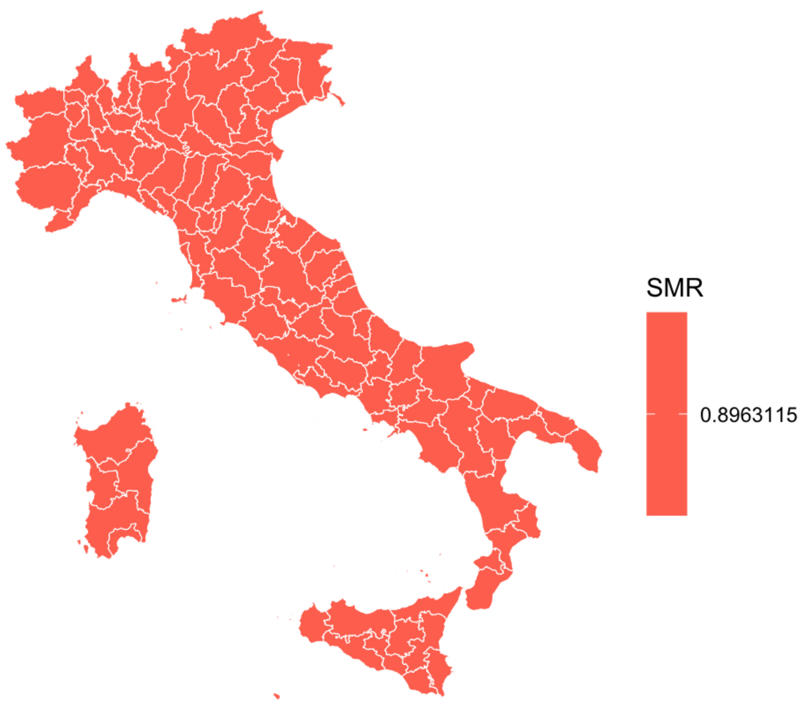
Priors:

$$\alpha \sim N(0, 5)$$

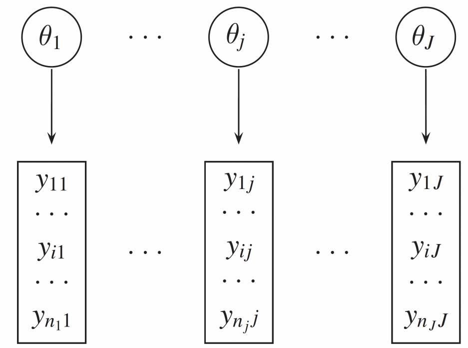
Likelihood:

$$\begin{aligned} y_i &\sim \text{Pois}(\mu_i) \quad i = 1, ..., N \\ \log(\mu_i) &= \log(E_i) + \alpha \end{aligned}$$

Complete pooling (from lab)



No pooling



- An *unpooled* model implies that the data are sampled from independent parameters.
- (θ_1,θ_2 etc.) drawn independently from e.g., $\theta_j \sim N(0, 1000)$.
- Parameter θ_j is different for each set of observations.
- No exchange of information between θ_j's.

No pooling (from lab)

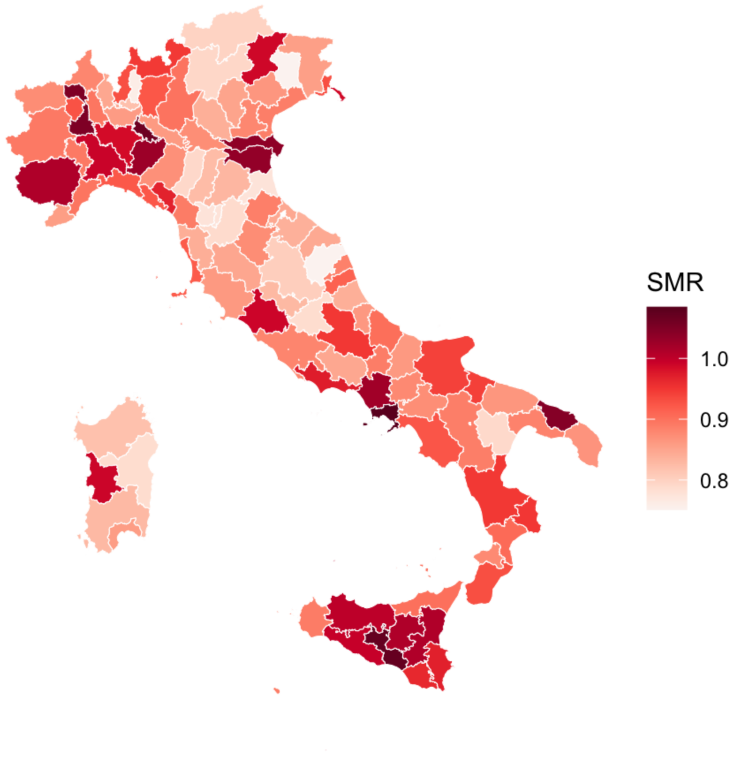
Priors:

$$\alpha_j \sim N(0,1) \quad j = 1, ..., N_p$$

Likelihood:

$$\begin{aligned} y_i &\sim \text{Pois}(\mu_i) \quad i = 1, ..., N \\ \log(\mu_i) &= \log(E_i) + \alpha_{j[i]} \end{aligned}$$

No pooling (from lab)



Hierarchical models

- Many real-world problems involve data grouped by source, geography, time, etc.
- Non-hierarchical models assume independence—often inappropriate.
- Hierarchical models:
  - Allow varying group-specific effects.
  - Improve estimates through sharing information (shrinkage).

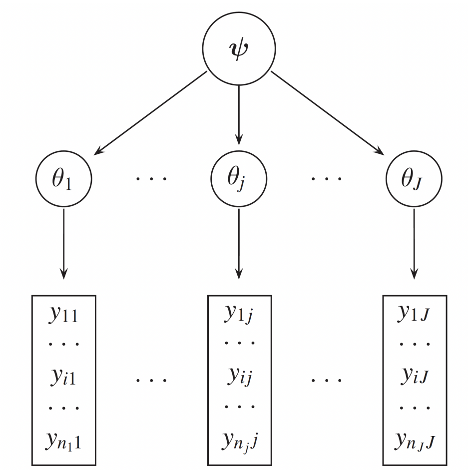
Hierarchical models

Exist on the continuum between two extreme cases:

- Complete pooling (one common parameter).
- Partial pooling!

- No pooling (independent parameters).

Partial pooling



- In a *partially pooled* or *multilevel* or *hierarchical* model, parameters are viewed as a sample from a distribution of parameters.

Partial pooling (from lab)

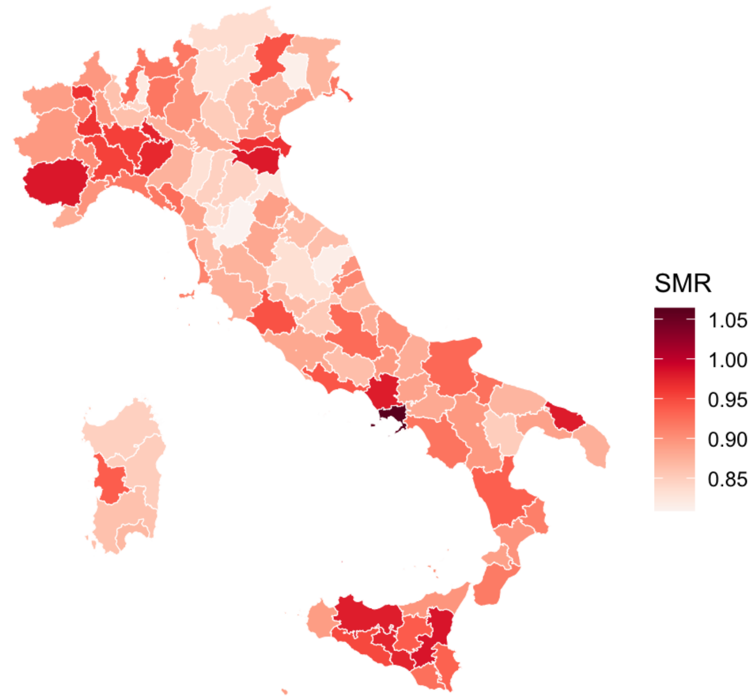
Priors:

$$\begin{aligned} \alpha &\sim N(0, 5), \\ \sigma_p &\sim N^+(1) \\ \theta_j &\sim N(0, \sigma_p^2) \quad j = 1, ..., N_p \end{aligned}$$

Likelihood:

$$\begin{aligned} y_i &\sim \text{Pois}(\mu_i) \quad i = 1, ..., N \\ \log(\mu_i) &= \log(E_i) + \alpha + \theta_{j[i]} \end{aligned}$$

Partial pooling (from lab)



Hierarchical (partial pooling) models

- Hierarchical models (partial pooling) fill in the continuum between the two extremes.
- They allow us to estimate models for each measurement block where each dataset is being fit to its own model.
- There is a higher level model, a *hierarchical model*, describing variability in parameters of those sub-models.

Comparison of pooling strategies

Pooling Type	Group estimates independent?	Shrinkage	Frequentist equivalent	Bayesian benefit
No Pooling	Yes	None	Fixed effects	None

Pooling Type	Group estimates independent?	Shrinkage	Frequentist equivalent	Bayesian benefit
Full Pooling	No (same for all)	Complete	No group effects	Simple prior
Partial Pooling	No (partially informed)	Adaptive	Random effects	Full posterior with hierarchy

Levels of hierarchy

- Data
- Process model (likelihood)
- Parameter model (priors)
- Hyperparameters (hyperpriors)

Partial pooling again

Priors:

$$
\begin{aligned}
\alpha &\sim N(0,5), \\
\sigma_p &\sim N^{+}(1) \\
\theta_j &\sim N(0,\sigma_p^2) \quad j = 1, ..., N_p
\end{aligned}
$$

Likelihood:

$$
\begin{aligned}
y_i &\sim \text{Pois}(\mu_i) \quad i = 1, ..., N \\
\log(\mu_i) &= \log(E_i) + \alpha + \theta_{j[i]}
\end{aligned}
$$

Comparing to frequentist random effects

- Frequentist random effects models are analagous in some ways to Bayesian hierarchical models, but with important differences.
- They allow for group-specific effects and account for intra-group correlation.
- However, they do not:
  - Provide a full probabilistic framework.
  - Incorporate prior information.

Comparing to frequentist random effects

- Bayesian hierarchical models:
  - Allow for uncertainty quantification at multiple levels.
  - Provide a coherent framework for model comparison and selection.
  - Enable the incorporation of prior knowledge and beliefs.
  - Allow for extensions in complexity on multiple levels (e.g., spatiotemporal models).

Bayesian advantages

- Handles complex multilevel and spatial hierarchies.
- Uses structured priors (CAR, GMRF) for spatial dependence.
- Supports flexible, non-Gaussian priors.
- Provides full uncertainty via posterior distributions.
- Easily incorporates expert knowledge through priors.
- Enables spatiotemporal and nonstationary extensions.
- Frequentist methods often limited or approximate.

Extending hierarchy

- Suppose there's another level of hierarchy, e.g.,
  - another level of geographic region below the regional level (e.g., city),
  - or individuals within regions.
  - We can extend the hierarchy by adding another level of parameters.
- For example, we can add a level for individuals within regions.
- This allows us to model individual-level variability while still accounting for regional effects.

Extending hierarchy

- The model can be expressed as:

$$
\begin{aligned}
y_i &\sim \text{Pois}(\mu_i) \quad i = 1, ..., N \\
\log(\mu_i) &= \log(E_i) + \alpha + \theta_{j[i]} + \phi_{k[i]} \\
\phi_k &\sim N(0, \sigma_\phi^2) \quad k = 1, ..., N_r
\end{aligned}
$$

- Here, $\phi_{k[i]}$ represents the individual-level effects, and σ_ϕ^2 is the variance of these effects.
- This allows us to capture individual-level variability while still accounting for regional effects.

Two-level IID hierarchical model

- A natural extension: data measured **repeatedly over time** within **multiple areas**.
- Example: daily respiratory admissions and $PM_{2.5}$ across J cities, over T years.
- Two sources of extra-Poisson variability:
  - **City-level effects** θ_j: stable differences between cities (e.g., demographics, baseline health).
  - **Year-within-city effects** γ_{jt}: year-specific deviations within each city (e.g., flu season severity, local policy changes).
- Both modelled as **IID Gaussian random effects**.

Two-level IID hierarchical model

$$\begin{aligned} y_{ijt} &\sim \text{Pois}(\mu_{ijt}) \quad i = 1, \dots, N_{jt},\ j = 1, \dots, J,\ t = 1, \dots, T \\ \log(\mu_{ijt}) &= \log(E_{ijt}) + \alpha + \beta x_{ijt} + \theta_j + \gamma_{jt} \end{aligned}$$

where:

- x_{ijt} is $PM_{2.5}$ concentration for observation i, city j, year t
- θ_j is the **city-level** random effect
- γ_{jt} is the **year-within-city** random effect

Two-level IID hierarchical model

Priors:

$$\begin{aligned} \alpha &\sim N(0, 5) \\ \beta &\sim N(0, 1) \\ \theta_j &\sim N(0, \sigma_\theta^2), \quad j = 1, \dots, J \\ \gamma_{jt} &\sim N(0, \sigma_\gamma^2), \quad j = 1, \dots, J,\ t = 1, \dots, T \\ \sigma_\theta &\sim N^+(0, 1) \\ \sigma_\gamma &\sim N^+(0, 1) \end{aligned}$$

- σ_θ^2 captures **between-city** variability.
- σ_γ^2 captures **within-city, between-year** variability.

Two-level IID hierarchical model

Directed Acyclic Graph (DAG):

```
Hyperpriors:      _                _
                  ↓                 ↓
Level 2:          _j          _{jt}
(city)
Level 1:            log( _{ijt})  ←   ,  , x_{ijt}
(obs)                ↓
Data:               y_{ijt}
```

- Each city j has its own **stable intercept** θ_j.
- Each city-year combination (j, t) has its own **transient deviation** γ_{jt}.
- Both shrink toward zero via their respective hyperpriors.

Two-level IID: what does shrinkage do here?

- Cities with **few observations** or **noisy data** are shrunk toward the global mean.
- Years with **extreme outcomes** are partially attributed to the year effect γ_{jt}, not the pollution coefficient β.
- This **protects** the estimate of β from confounding by city-level or year-level factors.
- The two variance components σ_θ^2 and σ_γ^2 are themselves estimated from the data — **adaptive regularisation**.

Two-level IID: variance decomposition

The total variability in log-risk can be decomposed as:

$$\mathrm{Var}(\log \mu_{ijt}) \approx \sigma_\theta^2 + \sigma_\gamma^2$$

Component	Symbol	Interpretation
Between-city variance	σ_θ^2	How much cities differ on average
Within-city, between-year variance	σ_γ^2	How much a city fluctuates year to year
Intraclass correlation (city)	$\dfrac{\sigma_\theta^2}{\sigma_\theta^2 + \sigma_\gamma^2}$	Proportion of variability due to city membership

- A **high ICC** suggests city-level factors dominate.
- A **low ICC** suggests year-to-year fluctuations are more important.

Model selection

- Model selection is the process of choosing between different models based on their performance on a given dataset.
- Frequentist:
  - Optimise different models on the **training set**.
  - Compare them on the **test set**.
- Bayesian:
  - Use **model evidence** $p(Data|Model_i)$
  - It tends to choose models which are just complex enough to model the data, but not overly complex (by which it avoids **overfitting**).

Information criteria

Information criteria are popular tools for model selection that provide a quantitative measure of the trade-off between model fit and complexity. The *lower* the value of IC, the better.

Common information criteria

- AIC (Akaike Information Criterion):

$$\text{AIC} = -2 * \ln(L) + 2k, L - \text{maximum value of likelihood}, k - \text{number of parameters}$$

- BIC (Bayesian Information Criterion):

$$\text{BIC} = -2\ln(L) + k\ln(n)$$

- Widely Applicable Information Criterion (WAIC):

$$\text{WAIC} = -2\sum_{i=1}^{n}(\ln(\hat{p}(y_i)) - \text{Var}(\ln(p(y_i))))$$

Getting ready for the lab

The lab for this session

- This lab will involve introducing hierarchical modelling using the `NIMBLE` modelling framework.
- During this lab session, we will:

1. Write a hierarchical model in `NIMBLE`;
2. Compare the model to more basic non-hierarchical models; and
3. Discuss the advantages of hierarchical models.

Questions?